

$$\begin{aligned}
& \left(-\frac{1}{16\pi^2} \left(-\frac{1}{8} g_2^4 y_e^{i1i2} \text{LF}_{3,0}[m_2] - \frac{1}{6} g_2^4 y_e^{i1i2} \text{LF}_{4,-1}[m_2] + \frac{8}{45} g_2^4 y_e^{i1i2} \text{LF}_{5,-2}[m_2] + \frac{1}{72} g_2^2 y_e^{i1i2} \right. \right. \\
& \left. \left. (-36 \overline{y_e^{pr}} y_e^{pr} + 5 \sum_p g_2^2) \text{LF}_{3,0}[m_1^p] - \frac{1}{48} g_2^2 y_e^{i1i2} (-24 \overline{y_d^{pr}} y_d^{pr} + \sum_p g_2^2) \text{LF}_{4,-1}[m_1^p] - \right. \right. \\
& \left. \left. \frac{2}{45} \sum_p g_2^4 y_e^{i1i2} \text{LF}_{5,-2}[m_1^p] + \frac{1}{24} g_2^2 y_e^{i1i2} (-36 \overline{y_d^{pr}} y_d^{pr} + 5 \sum_p g_2^2) \text{LF}_{3,0}[m_q^p] - \right. \right. \\
& \left. \left. \frac{1}{16} g_2^2 y_e^{i1i2} (-24 \overline{y_d^{pr}} y_d^{pr} + \sum_p g_2^2) \text{LF}_{4,-1}[m_q^p] - \frac{2}{15} \sum_p g_2^4 y_e^{i1i2} \text{LF}_{5,-2}[m_q^p] - \right. \right. \\
& \left. \left. \frac{1}{18} g_2^4 y_e^{i1i2} \text{LF}_{3,0}[\tilde{\mu}] - \frac{1}{12} g_2^4 y_e^{i1i2} \text{LF}_{4,-1}[\tilde{\mu}] + \frac{4}{45} g_2^4 y_e^{i1i2} \text{LF}_{5,-2}[\tilde{\mu}] + \right. \right. \\
& g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,0}[m_1, m_e^r] - 2 g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,-1}[m_1, m_e^r] + \\
& g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{4,1,-2}[m_1, m_e^r] - \frac{1}{4} g_1^4 y_e^{i1i2} \text{LF}_{2,2,-1}[m_1, m_e^{i2}] + \\
& \frac{1}{4} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,0}[m_1, m_1^p] - \frac{1}{2} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,-1}[m_1, m_1^p] + \\
& \frac{1}{4} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{4,1,-2}[m_1, m_1^p] + \frac{1}{32} g_1^2 y_e^{i1i2} (g_1^2 - g_2^2) \text{LF}_{2,2,-1}[m_1, m_1^{i1}] + \\
& \frac{3}{4} g_2^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,0}[m_2, m_1^p] - \frac{3}{2} g_2^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,-1}[m_2, m_1^p] + \\
& \frac{3}{4} g_2^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{4,1,-2}[m_2, m_1^p] + \frac{1}{32} g_2^2 y_e^{i1i2} (3 g_1^2 + g_2^2) \text{LF}_{2,2,-1}[m_2, m_1^{i1}] + \\
& \frac{2}{3} g_2^4 y_e^{i1i2} \text{LF}_{2,1,0}[m_2, \tilde{\mu}] + \frac{17}{12} g_2^4 y_e^{i1i2} \text{LF}_{2,2,-1}[m_2, \tilde{\mu}] - \frac{1}{3} g_2^4 y_e^{i1i2} \text{LF}_{3,1,-1}[m_2, \tilde{\mu}] - \\
& \frac{7}{3} g_2^4 y_e^{i1i2} \text{LF}_{3,2,-2}[m_2, \tilde{\mu}] + \frac{3}{2} g_2^4 m_2 y_e^{i1i2} \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] + g_2^4 y_e^{i1i2} \text{LF}_{3,3,-3}[m_2, \tilde{\mu}] - \\
& g_2^4 m_2^2 y_e^{i1i2} \text{LF}_{3,3,-2}[m_2, \tilde{\mu}] + g_2^4 y_e^{i1i2} \text{LF}_{4,2,-3}[m_2, \tilde{\mu}] - g_2^4 m_2^2 y_e^{i1i2} \text{LF}_{4,2,-2}[m_2, \tilde{\mu}] - \\
& \frac{1}{4} g_1^2 y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{2,2,0}[m_d^r, m_q^p] - \frac{1}{4} g_1^2 y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{2,2,0}[m_e^r, m_1^p] + \\
& \frac{1}{4} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,0}[m_e^r, \tilde{\mu}] - \frac{1}{8} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,-1}[m_e^r, \tilde{\mu}] + \\
& \frac{1}{2} g_1^4 y_e^{i1i2} \text{LF}_{2,1,0}[m_e^{i2}, m_1] - \frac{1}{2} g_2^2 y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,1,0}[m_1^p, m_e^r] + \\
& \frac{1}{4} g_1^2 y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,2,-1}[m_1^p, m_e^r] + \frac{1}{8} y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} (3 g_1^2 + 5 g_2^2) \text{LF}_{4,1,-1}[m_1^p, m_e^r] - \\
& \frac{1}{6} y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} (3 g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_1^p, m_e^r] + \\
& \frac{1}{2} \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} y_e^{i1i2} \text{LF}_{2,1,0}[m_1^p, m_1^s] - \frac{1}{2} \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} y_e^{i1i2} \text{LF}_{3,1,-1}[m_1^p, m_1^s] - \\
& \frac{1}{4} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,0}[m_1^p, \tilde{\mu}] + \frac{1}{8} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,-1}[m_1^p, \tilde{\mu}] + \\
& \frac{1}{16} g_1^2 y_e^{i1i2} (-g_1^2 + g_2^2) \text{LF}_{2,1,0}[m_1^{i1}, m_1] - \frac{1}{16} g_2^2 y_e^{i1i2} (3 g_1^2 + g_2^2) \text{LF}_{2,1,0}[m_1^{i1}, m_2] - \\
& \frac{1}{2} y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} (2 g_1^2 + 3 g_2^2) \text{LF}_{3,1,0}[m_q^p, m_d^r] + \frac{1}{4} g_1^2 y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{3,2,-1}[m_q^p, m_d^r] + \\
& \frac{3}{8} y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} (7 g_1^2 + 5 g_2^2) \text{LF}_{4,1,-1}[m_q^p, m_d^r] - \\
& \frac{1}{2} y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} (3 g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_q^p, m_d^r] + \\
& \frac{3}{2} \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} y_e^{i1i2} \text{LF}_{2,1,0}[m_q^p, m_q^s] - \frac{3}{2} \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} y_e^{i1i2} \text{LF}_{3,1,-1}[m_q^p, m_q^s] - \\
& \frac{1}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 + g_2^2) \text{LF}_{2,2,0}[m_q^p, m_u^r] + \frac{3}{2} g_2^2 \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \text{LF}_{3,1,0}[m_q^p, m_u^r] - \\
& \frac{1}{2} g_2^2 \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \text{LF}_{3,2,-1}[m_q^p, m_u^r] - \frac{3}{2} g_2^2 \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \text{LF}_{4,1,-1}[m_q^p, m_u^r] + \\
& \frac{1}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 - g_2^2) \text{LF}_{3,1,0}[m_u^r, m_q^p] + \frac{1}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 + g_2^2) \\
& \text{LF}_{3,2,-1}[m_u^r, m_q^p] + \frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_u^r, m_q^p] - \\
& \frac{1}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (3 g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_u^r, m_q^p] + \frac{1}{2} g_1^2 y_e^{i1i2} (g_1^2 + g_2^2) \text{LF}_{3,1,-1}[\tilde{\mu}, m_1] - \\
& \frac{3}{8} g_1^4 y_e^{i1i2} \text{LF}_{3,2,-2}[\tilde{\mu}, m_1] - \frac{3}{8} g_1^4 m_1^2 y_e^{i1i2} \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] - \\
& \frac{1}{12} g_1^2 y_e^{i1i2} (12 g_1^2 + 7 g_2^2) \text{LF}_{4,1,-2}[\tilde{\mu}, m_1] + \frac{1}{4} g_1^4 y_e^{i1i2} \text{LF}_{4,2,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{4} g_1^4 m_1^2 y_e^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{6} g_1^2 y_e^{i1i2} (3 g_1^2 + g_2^2) \text{LF}_{5,1,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{6} g_2^2 y_e^{i1i2} (9 g_1^2 + g_2^2) \text{LF}_{3,1,-1}[\tilde{\mu}, m_2] - \frac{85}{24} g_2^4 y_e^{i1i2} \text{LF}_{3,2,-2}[\tilde{\mu}, m_2] - \\
& \frac{3}{8} g_2^4 m_2^2 y_e^{i1i2} \text{LF}_{3,2,-1}[\tilde{\mu}, m$$